

Shyam S Balasubramanian

Bangalore, India | +91 9902265765 | shyam19sb@gmail.com | [linkedin.com/in/shyam19sb](https://www.linkedin.com/in/shyam19sb) | github.com/shyam19a

Summary

Engineering student with a strong foundation in **programming, problem-solving, and core computer science concepts**. Highly motivated, adaptable, and passionate about pursuing a career in **software engineering**. Interested in designing scalable solutions, building impactful applications, and continuously expanding technical expertise.

Education

- **Bachelor of Engineering in Electronics and Communication Engineering (ECE)**
Dayananda Sagar College of Engineering, Bangalore | CGPA: **8.17** | **2022 – 2026**
- **Class XII (CBSE)** | *The Amaatra Academy, Bangalore* | Percentage: **63.8%** | **2022**
- **Class X (ICSE)** | *St. Peter's School, Bangalore* | Percentage: **88.5%** | **2020**

Projects

CareerSync AI | *Personal Project*

- Developed a **web application** that leverages **AI** to compare resumes against job descriptions, highlighting matching/missing skills and generating a similarity score. Implemented **real-time skill extraction and matching logic** with a **FastAPI backend**, integrated seamlessly with a **React + Material-UI frontend**. Built an **interactive, responsive UI** providing instant feedback and actionable resume improvement suggestions.
Tech Stack: ReactJS, Material-UI, FastAPI, GitHub Actions/Pages

Energy Efficient Communication Protocol for WSN using FPGA

- Developed a hybrid clustering protocol combining **LEACH** and **HEED** to enhance energy efficiency in wireless sensor networks. Implemented the protocol on an **FPGA platform**, leveraging hardware parallelism for real-time execution, reduced power consumption, and improved network lifetime. The project demonstrated the feasibility of FPGA-based communication protocols for **IoT, smart agriculture, and healthcare applications**, bridging the gap between simulation models and practical deployment.

Certifications

- Introduction to Microsoft Azure Cloud Services - Microsoft
- Beginning C++ Programming - From Beginner to Beyond - Udemy
- CMOS Digital VLSI Design - NPTEL

Skills & Abilities

Programming Languages:	C++, JavaScript
CS Fundamentals:	Data Structures & Algorithms, OOPs, DBMS, Computer Networks, OS
Web Development:	HTML, CSS, Tailwind CSS, ReactJS, Node.js
Version Control:	Git, GitHub
Cloud:	Azure